Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. HEFTY: (A) arid (B) smooth (C) shapely (D) horizontal (E) lightweight

2. CRUCIAL: (A) inaccurate (B) unessential (C) uninspiring (D) preferential (E) reputable

3. RUFFLE: (A) enlarge (B) censor (C) avoid (D) calm (E) eject

4. CONGLOMERATE: (A) retract (B) darken (C) separate (D) reveal (E) stir

5. MONUMENTAL: (A) biased (B) vacant (C) trivial (D) ambiguous (E) ingenious

6. BRAWN: (A) frailness (B) silence (C) innocence (D) foolishness (E) sluggishness

7. DISLODGE: (A) amass (B) befriend (C) validate (D) solicit (E) entrench

8. ADVOCACY: (A) disrepute (B) opposition (C) ascendancy (D) justice (E) unconsciousness

9. COGNATE: (A) impartial (B) unrelated (C) unlikely (D) inconvenient (E) expendable

10. QUIXOTIC: (A) fortunate (B) practical (C) devoted (D) proud (E) kindly

11. WORKADAY: (A) superficial (B) competitive (C) extraordinary (D) confidential (E) deceptive

12. DISTENSION: (A) proximity (B) regularity (C) pleasure (D) contraction (E) repulsion

13. BEMUSE: (A) enlighten (B) pervade (C) savor (D) return promptly (E) make responsible

14. PROCLIVITY: (A) emptiness (B) recklessness (C) inactivity (D) negligence (E) disinclination

15. IMPECUNIOUS: (A) rich (B) strong (C) guilty (D) rude (E) hazardous

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Although it seems to have been a fixture of the square since the city's origin, the produce market actually opened only ----.

(A) enthusiastically (B) recently (C) frequently (D) illegally (E) graciously

17. The grave problems that afflicted him were the purely ---- ones of hunger, pain, and fatigue.

(A) physical (B) imaginary (C) irrelevant (D) hereditary (E) coincidental

18. Her ability to analyze issues fairly and competently has earned her the ---- of even her most ---- peers.

(A) solicitation. .unbiased
(B) attention. .sympathetic
(C) suspicion. .discerning
(D) reputation. .knowledgeable
(E) respect. .exacting

19. Until the mid-1800's, the occasional reports by European explorers of the existence of gorillas were generally disbelieved and swiftly ---- the realm of tall tales.

(A) extracted from (B) confused with (C) relegated to (D) compared to (E) exalted to

20. The historian's assertion that there is a ---- of verifiable information about nineteenth-century Native Americans is ---, for there exist many transcripts of authentic oral accounts of their activities.

(A) wealth. .fallacious
(B) shortage. .legitimate
(C) depletion. .irrefutable
(D) paucity. .erroneous
(E) surfeit. .implausible

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Public officials are now considering both the separate and additive effects on humans of lead, mercury, and cadmium, along with a variety of other chemicals in our atmosphere. Each of these materials, regarded separately or together, is known to interfere with several systems of the human body. The first of these, lead, poses a particular threat to large numbers of people. The quantity of lead that the average American absorbs suggests the probable emergence of as yet undiscovered disorders.

Even before 650 B.C., lead ore was mined on a small scale in order to extract the silver from it. With the invention of coinage, the demand for silver led to an enormous increase in the amount of available lead ore. At the time of the Roman Empire, the wealthy used elegant metal containers for food and drink. Since acidic wine leached copper from bronze or copper vessels, producing obvious toxic effects, containers came to be lined with lead, which did not cause vomiting and actually tasted good. In fact, however, lead is more toxic than copper, and yet the members of the Roman aristocracy added lead to their diets in more than a dozen ways.

During the second century A. D., the Roman aristocracy began to show signs of an inability to reproduce; it was gradually replaced by a new leadership, which adopted a similar way of life. This new aristocracy also failed to reproduce, and today we surmise that lead may have contributed to this failure. We know that lead causes miscarriages and sterility, and that human bones store ninety percent of the body's total burden of lead. The fact that poisonous levels of lead have been measured in the exhumed bones of Roman aristocrats should give us pause, for per capita use of lead by the Romans apparently reached only fifty percent of what it is in the United States today.

Our bodies absorb up to ten percent of the lead we eat, but up to fifty percent of the lead we breathe. In contrast to the atmosphere surrounding those Roman aristocrats, the atmosphere of our metropolitan areas is filled with lead, mostly from the exhaust of automobiles. This cloud of pollution seems to be increasing in area rather than density.

Much of the lead we take in is usually excreted in the urine, but a portion accumulates in the bones, replacing some of the calcium there. Under certain conditions, that lead may be released into the blood at toxic levels. Lead poisoning symptoms can appear in different parts of the body: the gastrointestinal tract, the central nervous system, the heart, the kidneys, and the liver. Apparently any amount of lead in the human body does some damage, but the effects of subclinical toxicity are difficult to identify. Uncertainty about the nature of low-level lead poisoning makes action to combat the problem frustratingly difficult to generate.

21. The passage is chiefly concerned with

(A) comparing the effect of lead with that of other atmospheric pollutants
(B) discussing the harm that can result from the absorption of lead
(C) suggesting how society can protect itself from chemical poisoning
(D) presenting a new theory to explain the fall of the Roman Empire
(E) showing the additive toxicity of lead with other environmental poisons

22. The passage suggests that by Roman times lead containers were widely used for which of the following reasons?

I. Lead was in abundant supply.
II. Lead imparted a pleasant taste to food and drink.
III. Lead was considered an antidote to copper poisoning.

(A) I only
(B) II only
(C) I and II only
(D) II and III only
(E) I, II, and III

23. The author's primary point in discussing the Roman aristocracy is best summarized by which of the following axioms?

(A) When in Rome, do as the Romans do.
(B) Power corrupts; absolute power corrupts absolutely.
(C) Every cloud has a silver lining.
(D) The past is prologue.
(E) No news is good news.

24. On the basis of the information in the passage, which of the following would do most to reduce the hazards currently associated with exposure to lead?

(A) Banning the use of metal containers for food and beverages
(B) Increasing the amount of calcium in daily diets
(C) Drinking large amounts of water
(D) Reducing the number of lead smelting plants
(E) Reducing the amount of lead in gasoline

25. The tone of the passage is best described as

(A) cautionary
(B) defensive
(C) cynical
(D) conspiratorial
(E) inspirational

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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The decision to use the Black man as a soldier against the Confederacy did not really grow out of any broad humanitarian resolve. General Halleck put the thing quite bluntly in a message to General Grant in March 1863. It was good policy, he said, to withdraw as many slaves from the South as possible; and an equally good policy, having withdrawn them, to use them to help win the war. If Grant found—as he undoubtedly would—that many of the people in his army objected, he must ride down their objections and see that this new policy was carried out.

This new policy was a good deal more far-reaching than Halleck dreamed. To accept the Black man as a soldier in the Union army was to state, in a backhanded but decisive way, that the base of membership in the American community had been immeasurably widened. Once widened, it could not again be narrowed. The war henceforth would be fought for this expanded base, even though some of the men who were most effectively fighting it had no idea that the base needed to be expanded.

Grant dutifully went to work, ordering his staff not only to organize Black regiments and render them effective, but also to remove prejudice against them. Removing that prejudice would not be easy, for some officers and enlisted men had vowed that they would throw down their arms and go home if Black men became soldiers. This anarchic opposition was quickly put down, partly because of Grant's orders and partly because of the unexpected intervention of a rather unlikely hero—lanky, dry-as-dust Lorenzo Thomas, adjutant general of the army, the paper shuffler from Washington. Part of Thomas' job was to speed the raising of Black regiments, and he took to this with crusty enthusiasm. He called troops together and warned them that Blacks fleeing from slavery were to be made welcome: "They are to be received with open arms; they are to be fed and clothed; they are to be armed." He was empowered, he added, to dismiss from the army "any man, be his rank what it may, whom I find maltreating the freedmen. This part of my duty I will most assuredly perform if any case comes before me."

One discontented army division was specially addressed by Thomas. Men who left the army because of the recruiting of Blacks, he warned, would be considered guilty of treason and would be shot, and there would be courts-martial for all who interfered with the program. The men talked it over around campfires afterward and decided to cooperate.

26. According to the passage, Halleck's order to Grant was motivated by considerations best described as

(A) capricious (B) ideological
(C) compassionate (D) vengeful
(E) pragmatic

27. According to the passage, in his message to Grant, Halleck anticipated which of the following difficulties?

(A) Reluctance of Black men to enlist
(B) Grant's objections to the proposal
(C) Strong disapproval from White soldiers
(D) Inadequate time to ready the new recruits for battle
(E) The public outcry against the policy

28. The author probably considers Lorenzo Thomas "a rather unlikely hero" (line 32) because Thomas

(A) was known to lack courage
(B) seemed a dull bureaucrat
(C) wielded no political clout
(D) was opposed to Halleck and Grant
(E) had little regard for army procedure

29. All of the following can be inferred from the passage to be measures taken by the Union army to accommodate Black soldiers EXCEPT

(A) providing the necessities of life
(B) training them for combat missions
(C) supplying them with needed weaponry
(D) granting them a voice in setting strategy
(E) punishing those who mistreated them

30. The author attributes such success as Halleck's policy had to

(A) military ingenuity
(B) urgency of need
(C) strong high-level support
(D) good fortune
(E) general good will

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. This book has neither merit nor distinction, and it all but ---- whatever prestige the author may once have had.

(A) challenges (B) maximizes (C) epitomizes (D) nullifies (E) encompasses

32. Far from exercising a counterinfluence to intellectual orthodoxy, the movement itself has become a center of ----.

(A) conformity (B) notoriety (C) enmity (D) theology (E) idealism

33. People, being prisoners of ----, have made falling a basic metaphor for failure, from the fall of the stock market to the fall of a government.

(A) politics (B) gravity (C) greed (D) turbulence (E) emotion

34. The movie avoids many of the ---- of the novel, bringing into sharper focus characters who originally were amiable or disagreeable ----.

(A) subtleties. .monsters
(B) excellences. .stereotypes
(C) allusions. .individuals
(D) particulars. .nonentities
(E) shortcomings. .blurs

35. Even the most insulated and ---- government has to be concerned today about the popular mind; some systems may be less ---- than others, but at some point all must pay attention.

(A) parochial. .vociferous
(B) corrupt. .theoretical
(C) arbitrary. .sensitive
(D) obscure. .refined
(E) vibrant. .responsive

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. GRAPE : VINE :: (A) wire : pole (B) snail : shell (C) cherry : pie (D) pumpkin : seed (E) apple : tree

37. SUIT : OVERCOAT :: (A) wig : hairdo (B) finger : glove (C) parasol : sun (D) shoes : galoshes (E) uniform : costume

38. BAT : MAMMAL :: (A) pine : tree (B) pup : seal (C) butterfly : insect (D) starfish : clam (E) ram : sheep

39. PLAGIARIZE : WORDS :: (A) embezzle : funds (B) imagine : visions (C) renovate : repairs (D) paint : colors (E) edit : publications

40. ENDURE : SURVIVOR :: (A) condemn : culprit (B) applaud : performer (C) evade : guardian (D) excel : imitator (E) compete : rival

41. PROGRAM : DRAMA :: (A) scale : map (B) agenda : meeting (C) footnote : article (D) amendment : bill (E) ticket : performance

42. VENDETTA : REVENGE ::
(A) exploration : discovery
(B) excitement : frenzy
(C) investment : deficit
(D) mistake : correction
(E) insomnia : nightmare

43. PREMISE : ARGUMENT ::
(A) location : filming
(B) profit : contract
(C) foundation : edifice
(D) presentation : package
(E) referee : dispute

44. SUBJUGATE : OBEDIENCE ::
(A) think : intelligence (B) arrest : illegality (C) frighten : apprehension (D) annoy : mistake (E) describe : event

45. TEMPER : ACTION :: (A) breathe : air (B) soothe : pleasure (C) soften : blow (D) operate : machine (E) shock : audience

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

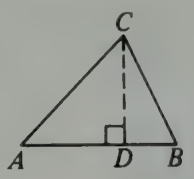
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $(2+3)(n+5)=30$, then $n=$

(A) 0 (B) 1 (C) 2 (D) 5 (E) 6

	Arthur	Bob
Round 1	15	47
Round 2	11	15
Round 3	47	49
Round 4	60	3
Round 5	17	34
Total	150	148

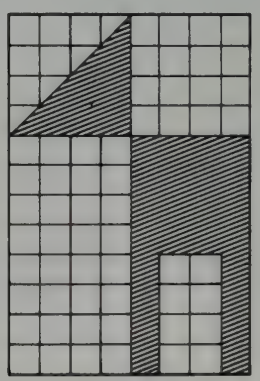
2. Arthur and Bob played five rounds of a card game. The table above shows their scores in the individual rounds. By how many points was Bob leading Arthur at the end of the first three rounds?

(A) 2 (B) 4 (C) 32 (D) 38 (E) 48

---

3. If $\dfrac{3^2+3^2+3^2}{3^x}=3$, then $x=$

(A) 2
(B) 3
(C) 9
(D) 27
(E) 81



4. In the grid of squares above, each small square has side of length 1. What is the sum of the areas of the shaded regions?

(A) 48 (B) 40 (C) 32 (D) 24 (E) 16

---

5. If the first and last digits are interchanged in each of the following numbers, which will yield the number with the least value?

(A) 4,321 (B) 3,241 (C) 2,431
(D) 4,231 (E) 3,421

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6. If $\frac{2x}{3} = 8$, then $4x =$

(A) 12 (B) 16 (C) 24 (D) 32 (E) 48

Card	Entry
A	N
B	$N \times 2$
C	$(N \times 2) \div 3$
D	$[(N \times 2) \div 3] + 15$
E	$[(N \times 2) \div 3] + 15 - 1$

7. According to the table above, which card has the entry with the greatest value when $N = 15$?

(A) *A* (B) *B* (C) *C* (D) *D* (E) *E*

---

8. If x and y are positive numbers, $xy = 24$, and $\frac{x}{y} = \frac{3}{2}$, what is the value of x?

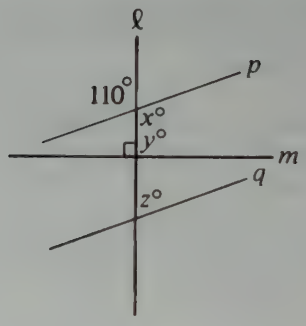
(A) 3
(B) 4
(C) 6
(D) 8
(E) 12

9. If the lengths of two sides of an isosceles triangle are 5 and 9, what is the length of the third side?

(A) 4 (B) 5 (C) 7 (D) 9
(E) It cannot be determined from the information given.

10. If $x = 9.000001$, then $\frac{x^2 - 7x + 6}{x - 1}$ rounded to the nearest whole number is

(A) 3
(B) 6
(C) 8
(D) 15
(E) 24



11. In the figure above, ℓ is perpendicular to m and p is parallel to q. What is the value of $x + y + z$?

(A) 180 (B) 220 (C) 230
(D) 270 (E) 310

---

12. Jean worked x hours and earned \$20. Pam worked y hours and earned \$40. If they both earn the same hourly wage and $x + y = 15$, how many hours did Jean work?

(A) 2
(B) 4
(C) 5
(D) 8
(E) 10

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13. The average (arithmetic mean) of Pat's scores for 3 rounds of golf was 75. If the average of her scores of the first 2 rounds was also 75, what was her score for the third round?

(A) 70
(B) 75
(C) 80
(D) 85
(E) It cannot be determined from the information given.

14. If $5m+4n=14$ and $4m+3n=11$, then $7(m+n)=$

(A) 7
(B) 14
(C) 21
(D) 28
(E) 35

15. If 5 people share a cash prize equally, what percent of the prize do 2 of them together receive?

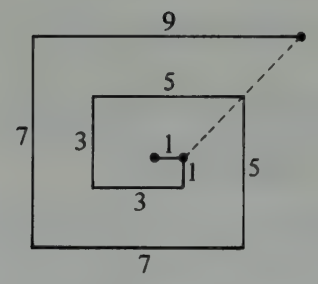
(A) 5% (B) 10% (C) 20%
(D) 25% (E) 40%

16. Two circles with radii r and $r+2$ have areas that differ by 8π. What is the radius of the larger circle?

(A) 1
(B) 2
(C) 3
(D) 8
(E) 9

17. For how many two-digit positive numbers will doubling the tens' digit yield a two-digit number that is double the original number?

(A) None (B) One (C) Two
(D) Three (E) Four



18. The figure above is formed by connecting perpendicular line segments that have lengths as shown. What is the length of the dashed line segment?

(A) 5 (B) $2\sqrt{7}$ (C) $4\sqrt{2}$
(D) $\sqrt{34}$ (E) $\sqrt{41}$

---

19. The operation $\odot$ is defined for all numbers s and t by the equation $s \odot t = st+1$. For example, $4 \odot 5 = 4 \cdot 5 + 1 = 21$. If $s \neq 0$ and r is a number such that $s \odot r = 1$, then $r=$

(A) $-\frac{1}{s}$
(B) 0
(C) $-s+1$
(D) $\frac{1}{s}$
(E) $s+1$

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Questions 20-21 refer to the following game.

A computer generates "words," and points are assigned as shown in the table below each time any of the four letter pairs given appears in a word.

Letter Pair	Number of Points
"AA"	10
"AB"	4
"BA"	3
"BB"	5

For example, the word BABT is assigned 3 points for "BA" and 4 more for "AB," making a total of 7 points.

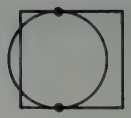
20. Which of the following words would be assigned the most points?

(A) SABBAQ
(B) BUBBLZ
(C) TABBYL
(D) BARNAB
(E) AARDVA

21. If 12 points are assigned to a certain word, which of the following statements must be true?

I. AA is not in the word.
II. AB and BA are both in the word.
III. If BB is in the word, then AB is in the word.

(A) I only (B) II only (C) III only
(D) I and III only (E) I, II, and III



22. In the figure above, the area of the square with two sides tangent to the circle is $4a^2\pi$. What is the area of the circle?

(A) $a\pi^2$ (B) $a^2\pi^2$ (C) $a^2\pi$ (D) $2a^2\pi$
(E) It cannot be determined from the information given.

---

23. The sum of n consecutive positive integers is always an even integer if n is a multiple of

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

24. The ratio of John's allowance to Mary's is 2 to 5. The ratio of John's allowance to Bill's is 3 to 7. What is the ratio of Mary's allowance to Bill's ?

(A) 15 to 14
(B) 14 to 15
(C) 10 to 21
(D) 35 to 6
(E) 6 to 35

25. If the volume of a cube is 1 cubic centimeter, then the distance from any vertex to the center point inside the cube is

(A) $\frac{1}{2}$ cm

(B) $\frac{\sqrt{2}}{2}$ cm

(C) $\sqrt{2}$ cm

(D) $\frac{\sqrt{3}}{2}$ cm

(E) $\sqrt{3}$ cm

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. SINCERE: (A) puzzled (B) depressed (C) repetitive (D) dishonest (E) unimaginative

2. ANCHOR: (A) release from restraint (B) ease anxiety (C) gauge accurately (D) board carefully (E) prevent spoilage

3. QUACK: (A) wealthy inventor (B) popular politician (C) qualified physician (D) well-known writer (E) entertaining actor

4. IRREPARABLE: (A) amiable (B) mendable (C) grateful (D) confusing (E) divisible

5. BARBARISM: (A) acceptable diction (B) pious excuse (C) liberating action (D) vague definition (E) antagonistic gesture

6. DITHER: (A) repeat (B) blame (C) act with assurance (D) respond as expected (E) record with accuracy

7. COPIOUS: (A) selfish (B) unholy (C) spoiled (D) outrageous (E) sparse

8. SUBTLE: (A) gross (B) brave (C) convincing (D) determined (E) overcautious

9. ACCOLADE: (A) sobriety (B) debate (C) reproach (D) disorder (E) refreshment

10. INVETERATE: (A) nervous (B) unthinking (C) ambiguous (D) naturally skeptical (E) readily changeable

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Explanations given to the patient by the anesthetist prior to surgery often ---- anxiety and ---- the need for analgesics.

(A) intensify. .counteract
(B) parallel. .assume
(C) explore. .prepare
(D) relieve. .reduce
(E) ignore. .preclude

12. The jellyfish's slow pulsing action propels it in a graceful, seemingly ---- drift, but its tentacles contain a poison potent enough to stun a swimming human.

(A) sinister (B) rhythmic (C) murky (D) harmless (E) patient

13. The production's slight romantic appeal depends on its two stars, who are ---- to watch even when they are delivering ---- speeches.

(A) overbearing. .innocent
(B) farcical. .complex
(C) attractive. .creative
(D) impossible. .dull
(E) pleasant. .empty

GO ON TO THE NEXT PAGE

4

14. The murderous capriciousness of Nero infused the Roman Empire with such pervasive ---- that not even the most ---- citizens felt safe.

(A) distaste. .vulgar
(B) paranoia. .suspicious
(C) enthusiasm. .apathetic
(D) unease. .oppressed
(E) panic. .distinguished

15. His inclination to succumb to flattery made him ---- to the ---- of people who wished to take advantage of him.

(A) immune. .predilection
(B) prejudicial. .intentions
(C) susceptible. .cajolery
(D) resistant. .blandishments
(E) amenable. .rejection

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. SPILL : POUR :: (A) lose : find
(B) stutter : hear (C) neglect : care
(D) stumble : walk (E) ponder : forget

17. INTRUDER : PRIVACY :: (A) blot : ink
(B) ripple : calm (C) noise : clamor
(D) animal : forest (E) hermit : solitude

18. FADED : COLOR :: (A) humid : moisture
(B) livid : anger (C) pungent : smell
(D) muted : sound (E) cloying : taste

19. PSEUDONYM : NAME :: (A) disguise : identity
(B) motive : purpose (C) talent : stardom
(D) mission : dedication (E) prestige : fame

20. EXCLAIM : UTTER :: (A) suppose : describe
(B) conceal : destroy (C) hurl : throw
(D) blink : stare (E) adore : spoil

21. FLEDGLING : BIRD :: (A) sty : pig
(B) kitten : cat (C) fodder : cow
(D) school : fish (E) stallion : horse

22. OSTENTATIOUS : APPAREL ::
(A) proud : character
(B) palatable : food
(C) bombastic : language
(D) voluminous : correspondence
(E) bedecked : flowers

23. INDOLENT : WORK :: (A) inquisitive : learn
(B) judgmental : accuse (C) blithe : laugh
(D) confident : win (E) modest : boast

24. REPAST : NOURISHES ::
(A) theory : verifies
(B) severance : attaches
(C) remedy : diagnoses
(D) inoculation : immunizes
(E) exhaustion : refreshes

25. NEFARIOUS : VIRTUE ::
(A) remorseless : pity
(B) ingenuous : youth
(C) altruistic : money
(D) melancholy : sympathy
(E) seductive : motivation

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Science fiction gives expression to thoughts and emotions that I think it valuable to entertain. It is sobering and cathartic to remember, now and then, our collective smallness, our apparent isolation, the seeming indifference of nature, the slow biological, geological, and astronomical processes that may, in the long run, make many of our hopes (and some of our fears) ridiculous.

Science fiction stories of this type may explain the hardly disguised political rancor I detected in one article on science fiction. The insinuation was that those who read or wrote it were probably hostile to the state. What lurks behind such a hint is probably something like this. If we were all on board ship and there were trouble among the stewards, I can just conceive the chief steward looking with disfavor on anyone who stole away from the fierce debates in the pantry to take a breather on deck. For up there one would taste the salt, would see the vastness of the water, would remember that the ship had a whither and a whence. What had seemed, in the hot, lighted rooms down below, to be the scene for a political crisis would appear once more as a tiny eggshell moving rapidly through an element in which humanity cannot survive. It would not necessarily change the steward's convictions about the rights and wrongs of the dispute down below, but it would probably show them in a new light.

26. The author asserts that readers and writers of science fiction can most accurately be described as

(A) seekers after an implausible dream
(B) those who view humanity in the largest perspective
(C) refugees from a harsh reality
(D) those who are best qualified to make moral judgments
(E) enemies of the established order

27. The author uses the story of the ship to do which of the following?

(A) Introduce an unpopular idea
(B) Revoke a belief formerly held by the author
(C) Provide an example of a confusing problem
(D) Dramatize a typical science fiction episode
(E) Illustrate a belief held by the author

28. Part of what the author means in saying that the ship has "a whither and a whence" (line 21) is that

(A) individuals in a state should not argue
(B) politicians have sought and will seek office because they desire power
(C) we lack a clear conception of our own reason for existence
(D) we should not forget that the world has a past and a future
(E) what has occurred and what will occur in society is inevitable

GO ON TO THE NEXT PAGE

4

There is such an incalculable amount of sand—wet and dry—in the world that geologists have had a hard time accounting for it. Sandstone is a minor source; most sand starts as tiny crystals of quartz which break off granite and other hard igneous rocks. Gypsum is still another source; some of the dunes at White Sands National Monument in New Mexico are almost pure gypsum from a dried-up lake bed.

Wind-borne sand particles scud along the ground, colliding with each other, bouncing off obstructions and wearing off their rough irregularities. Eventually, smoothed and rounded, they approach a perfectly spherical shape—and may keep it, without further wearing, for millions of years. It was once believed that sand grains were rounded while washing down river beds, but laboratory experiments showed that they are too lightweight to abrade each other in water. A cube of quartz a fiftieth of an inch across, it was estimated, would have to be transported by water a distance equivalent to 50 times around the world before it became fully rounded, but wind abrasion would round it off 100 to 1,000 times more rapidly. Thus it is evident that most of the rounded sand grains in the world have been exposed to wind abrasion at one time or another.

29. The author's reference to White Sands National Monument does which of the following?

(A) Illustrates a general statement
(B) Resolves an apparent paradox
(C) Raises a new question
(D) Emphasizes a contrast
(E) Proves a theory

30. The results of the laboratory experiments were significant because they

(A) proved that moving water is the sole agent in the creation of sand
(B) revised current beliefs that scientists held about the size of sand grains
(C) enabled elaborate calculations to take place
(D) called into question a previously accepted model
(E) provided controversial evidence of the makeup of sand

31. The passage indicates that grains of sand assume their final shape primarily through which of the following?

(A) Debilitation (B) Sedimentation
(C) Fragmentation (D) Compounding
(E) Weathering

32. The passage is most probably an excerpt from which of the following?

(A) An informal article written for a general audience
(B) A proposal for a complex research project
(C) A college textbook that explains the laws of physics
(D) A report by a scientist who made a dramatic discovery
(E) An impressionistic memoir by a distinguished geologist

GO ON TO THE NEXT PAGE

(This passage was written in 1855.)

It would seem to us that the late edition of Margaret Fuller's *Woman in the Nineteenth Century* has been unduly thrust into the background by less comprehensive and candid productions on the same subject. Notwithstanding certain defects of taste and a sort of vague spiritualism and grandiloquence that belong to all but the very best American writers, the book is a valuable one. There is no exaggeration of woman's moral excellence or intellectual capabilities; no injudicious insistence on her fitness for this or that function hitherto engrossed by men; but a calm plea for the removal of unjust laws and artificial restrictions, so that the possibilities of her nature may have room for development.

So far as the difference in date allows, there is a striking similarity in the trains of thought of Margaret Fuller and of Mary Wollstonecraft, who wrote *The Rights of Women* sixty years before *Woman in the Nineteenth Century* was written. Both writers communicate a strong understanding of the subject; but Margaret Fuller's mind is like some regions of her own American continent, where you are constantly stepping from the sunny clearings into the mysterious twilight of the tangled forest—she often passes in one breath from forcible reasoning to dreamy vagueness. Mary Wollstonecraft, on the other hand, is nothing if not rational; her grave pages are lit up by no ray of fancy. Margaret Fuller is more of the literary woman, who would not have been satisfied without intellectual production; Mary Wollstonecraft, we imagine, wrote not at all for writing's sake, but from the pressure of other motives.

33. It can be inferred from the passage that which of the following can be found in *Woman in the Nineteenth Century*?

I. Reasoned arguments
II. Lucid assessments
III. Hazy discourses

(A) I only
(B) II only
(C) I and II only
(D) I and III only
(E) I, II, and III

34. In context, the word "engrossed" (line 11) is best interpreted to mean

(A) avoided
(B) pondered
(C) monopolized
(D) collected
(E) mismanaged

35. According to the author, Margaret Fuller and Mary Wollstonecraft are alike in that they

(A) have similar literary styles
(B) belonged to the same generation
(C) became writers for similar reasons
(D) have both been unduly overlooked
(E) express many of the same ideas

GO ON TO THE NEXT PAGE

4

At midnight it was so dark that the most practiced eye could barely pierce the gloom. But a faint drip of oars struck the ears of the Spaniards as they watched from the decks. A few moments afterward, the sea became suddenly luminous, and six flaming galleons appeared at a slight distance, bearing steadily down upon them before the wind and tide.

The Spanish captain, Medina-Sidonia, and most of his sailors had engaged in King Philip's siege of Antwerp three years before. They realized with horror that the flaming devil-ships designed by Gianibelli on that earlier occasion were now turned against them, rending sky and ocean in their wake. The Spaniards knew that the famous engineer was at that moment in the employ of Queen Elizabeth and the English navy.

In a moment, panic spread with contagious rapidity among Medina-Sidonia's men. There were shouts throughout the fleet, "The fire ships of Antwerp, the fire ships of Antwerp!" In an instant, every cable was cut and every galleon made frantic attempts to escape imminent destruction. The confusion was beyond description. Four of the clumsiest ships became entangled with each other. Two others were consumed by the flaming vessels. Medina-Sidonia, who had been warned by King Philip's spies that some such artifice might be attempted, and who had, early that morning, sent out a party of sailors to search for indications of the scheme, was not dismayed. He commanded every ship to return to its post and await further orders when the danger was past. But it was useless in that moment of unreasonable panic to issue commands.

The despised Mantuan had met with many rebuffs from the Spaniards at King Philip's court. Owing to official incredulity, he had been but partially successful in engineering the Spanish offensive at Antwerp. But now he had his revenge, for by the mere terror of his reputation, he inflicted more damage on Philip's fleet than had been accomplished by all of Queen Elizabeth's seadogs: Howard, Drake, Hawkins, and Frobisher.

While darkness lasted, the uproar continued. When Monday morning dawned, several of the huge Spanish vessels lay disabled, while the rest of the once-mighty fleet drove on toward the Flemish coast. The threatened gale had begun to blow, and those awkward riders of the storm floundered in the heavy seas. Meanwhile, the small but mobile English fleet was all astir to pursue the Spaniards, now drifting into the North Sea.

36. In describing the assault of the English fire ships, the author does NOT explain why

(A) Medina-Sidonia was not surprised by the English tactics
(B) the Spanish sailors were terrified
(C) it was useless for Medina-Sidonia to issue commands
(D) a search party had been sent out that morning
(E) Medina-Sidonia had failed to forewarn his men

37. It can be inferred from the passage that the "despised Mantuan" (line 34) is

(A) the English admiral
(B) Medina-Sidonia
(C) Gianibelli
(D) the captain of one of the "devil-ships"
(E) the king of Spain

38. The phrase "riders of the storm" (line 48) refers to

(A) the English sailors
(B) the Spanish sailors
(C) the English ships
(D) the Spanish ships
(E) Gianibelli's sailors

39. All of the following can be inferred about Medina-Sidonia EXCEPT that he

(A) suspected an assault by the English fire ships
(B) ordered his fleet to surrender
(C) remained calm in the face of attack
(D) was in the employ of King Philip
(E) had been present at the siege of Antwerp

40. According to the passage, the primary cause of the rout of the Spanish fleet was the

(A) Spanish sailors' lack of experience
(B) bravery of the English seadogs
(C) fear inspired by Gianibelli's reputation
(D) rigors of a storm at sea
(E) unreasonable orders of Medina-Sidonia

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

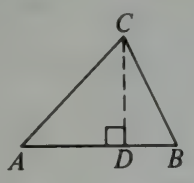
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

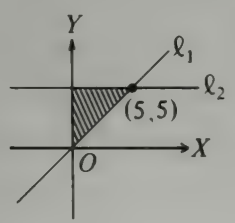
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1. What number increased by 8 equals 3 times the number?

   (A) 2 (B) 3 (C) 4 (D) 5 (E) 6

2. If it was 3:15 four hours ago, what time will it be $\frac{1}{2}$ hour from now?

   (A) 3:45 (B) 6:45 (C) 7:15
   (D) 7:45 (E) 11:45



3. In the figure above, if ℓ_2 is parallel to the X-axis, which of the following points lies in the shaded region?

   (A) (2, 4) (B) (−2, 4) (C) (2, 6)
   (D) (4, 2) (E) (6, 2)

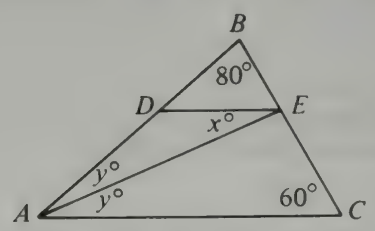
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4. Of the following, the closest approximation to $\frac{0.512(198.7)}{2.01}$ is

   (A) 5 (B) 25 (C) 50 (D) 100 (E) 2,500

5. Jeff is taller than Kim, but he is shorter than Mary. If j, k, and m are the heights in inches of Jeff, Kim, and Mary, respectively, which of the following is true?

   (A) $j < k < m$
   (B) $k < j < m$
   (C) $k < m < j$
   (D) $m < j < k$
   (E) $m < k < j$



Note: Figure not drawn to scale.

6. In $\triangle ABC$ above, if $DE \parallel AC$, then $x =$

   (A) 70 (B) 50 (C) 40 (D) 30 (E) 20

---

7. On a pool table there are 15 balls, each labeled with a different integer from 1 to 15 inclusive. If 3 even-numbered balls are removed, what fraction of the remaining balls are even numbered?

   (A) $\frac{4}{7}$ (B) $\frac{1}{2}$ (C) $\frac{5}{12}$ (D) $\frac{1}{3}$ (E) $\frac{1}{4}$

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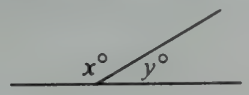
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

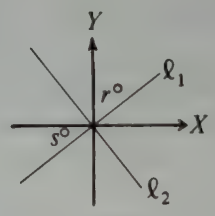
	Column A	Column B
	$34 - 2n = 6$	
8.	n	20

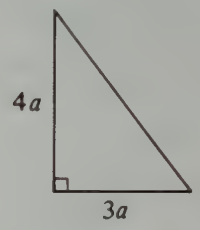
One face of a solid cube is white and the remaining faces are blue.

9.	The number of faces of the cube that are blue	4
10.	$\sqrt{35} + \sqrt{63}$	$6 + 8$
11.	The number of days in x years	y days

x is an odd integer and $6 < x < 9$.

12.	$x + 1$	8

	Column A	Column B
	 $\ell_1 \perp \ell_2$	
13.	$r + s$	90
14.	150×250	200×200



15.	$(3a)^2 + (4a)^2$	$(5a)^2$
	$n > 0$	
16.	$\frac{n+2}{3}$	$\frac{n+4}{6}$
	$x > 1$	
17.	The average speed when x kilometers is traveled in 1 hour	The average speed when 1 kilometer is traveled in x hours

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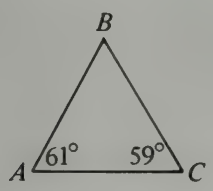
SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B
	In $\triangle ABC$, AC is 5 and CB is 8.	
18.	Area $\triangle ABC$	20

	Column A	Column B
	For all positive x, $\triangle x = \frac{x}{2}$. For all negative x, $\triangle x = x^2$.	
19.	$\triangle -4$	$\triangle 32$

	Column A	Column B
	$-\frac{1}{2} = \frac{1}{2}x$	
20.	$-x$	$-\frac{1}{4}$

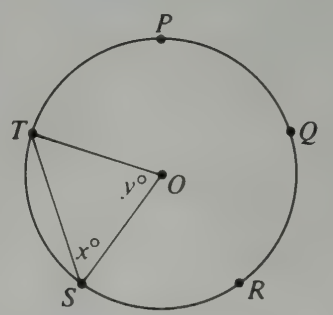
	Column A	Column B
		
21.	Length of AC	Length of BC

	Column A	Column B
	n is a positive integer and $0 < x < 1$.	
22.	$\frac{n}{x}$	n

	Column A	Column B
	The average (arithmetic mean) of x and y is 4 and $x - y + 2 = 0$.	
23.	x	3

	Column A	Column B
	p, q, r, s, and t are all positive integers. $p < q < r < s < t$ $p \times q = 3$ and $s \times t = 50$	
24.	r	4

	Column A	Column B
	Of 100 workers, exactly 90 percent pick pears, exactly 80 percent pick apples, and exactly 60 percent pick peaches.	
25.	The number of workers who pick both apples and peaches	40



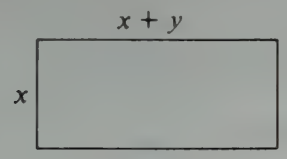
	Column A	Column B
	Points P, Q, R, S, and T are equally spaced on the circumference of the circle with center O.	
26.	x	y

	Column A	Column B
	a, b, and c are positive.	
27.	$a + b + c$	$\frac{1}{a + b + c}$

GO ON TO THE NEXT PAGE

5

Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.



28. The rectangle above has width x and length $x+y$. What is its perimeter in terms of x and y?

(A) x^2+xy
(B) $4x+2y$
(C) $4x+y$
(D) $2x+2y$
(E) $2x+y$

---

29. If x and y are positive integers and $x+2y=9$, then the greatest possible value of x is

(A) 3 (B) 4 (C) 6 (D) 7 (E) 9

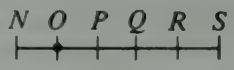
30. If x and y are negative numbers, which of the following must always be negative?

I. $x+y$
II. $x \cdot y$
III. $x-y$

(A) I only
(B) I and II only
(C) I and III only
(D) II and III only
(E) I, II, and III

31. Alice has saved \$39 in 6 weeks. If she continues to save at the same weekly rate, in how many more weeks will her total savings be \$104?

(A) 22
(B) 16
(C) 14
(D) 12
(E) 10



32. In the figure above, NS is divided into five equal segments as shown. A circle, not shown, with center O and radius $\frac{1}{4}$ the length of NS will intersect NS between

(A) N and O
(B) O and P
(C) P and Q
(D) Q and R
(E) R and S

$$\begin{array}{r} P\,Q \\ +\,Q\,P \\ \hline T\,P\,T \end{array}$$

33. Each of the three letters in the sum above represents a different digit. What is the value of P?

(A) 5 (B) 4 (C) 3 (D) 2 (E) 1

---

34. $\frac{1}{10^{19}} - \frac{1}{10^{20}} =$

(A) $\frac{1}{10}$ (B) $\frac{9}{10^{20}}$ (C) $\frac{1}{10^{20}}$

(D) $-\frac{9}{10^{20}}$ (E) $-\frac{1}{10}$

35. The average (arithmetic mean) of three integers p, r, and 6 is 8. Which of the following could NOT be the value of the product pr?

(A) 17 (B) 18 (C) 77 (D) 80 (E) 81

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test Form Codes 4E and 4V

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. E	1. D	1. B	1. C
2. B	2. A	2. D	2. D
3. D	3. C	3. A	3. A
4. C	4. B	4. C	4. C
5. C	5. A	5. D	5. B
6. A	6. C	6. E	6. E
7. E	7. E	7. B	7. D
8. B	8. A	8. C	*8. B
9. B	9. C	9. E	*9. A
10. B	10. E	10. A	*10. B
11. C	11. D	11. D	*11. D
12. D	12. D	12. C	*12. C
13. A	13. E	13. B	*13. C
14. E	14. E	14. C	*14. B
15. A	15. C	15. E	*15. C
16. B	16. D	16. C	*16. A
17. A	17. B	17. E	*17. A
18. E	18. D	18. C	*18. D
19. C	19. A	19. B	*19. C
20. D	20. C	20. A	*20. A
21. B	21. B	21. D	*21. B
22. C	22. C	22. B	*22. A
23. D	23. E	23. C	*23. C
24. E	24. D	24. A	*24. C
25. A	25. A	25. D	*25. D
26. E	26. B		*26. B
27. C	27. E		*27. D
28. B	28. D		28. B
29. D	29. A		29. D
30. C	30. D		30. A
31. D	31. E		31. E
32. A	32. A		32. C
33. B	33. E		33. D
34. E	34. C		34. B
35. C	35. E		35. B
36. E	36. E		
37. D	37. C		
38. C	38. D		
39. A	39. B		
40. E	40. C		
41. B			
42. A			
43. C			
44. C			
45. C			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

$$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 278. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 278. Enter this figure on the worksheet.

SAT-SCORING WORKSHEET

FORM CODES 4E and 4V

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Codes 4E and 4V

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	450	590
84	780		39	440	580
83	770		38	440	570
82	760		37	430	560
81	750		36	420	550
80	740		35	420	540
79	730		34	410	530
78	720		33	410	530
77	710		32	400	520
76	700		31	390	510
75	690		30	390	500
74	680		29	380	490
73	680		28	370	480
72	670		27	370	470
71	660		26	360	460
70	650		25	350	450
69	650		24	350	440
68	640		23	340	430
67	630		22	330	420
66	620		21	330	410
65	610		20	320	410
64	610		19	310	400
63	600		18	310	390
62	590		17	300	380
61	580		16	290	370
60	580	800	15	280	360
59	570	780	14	280	350
58	560	760	13	270	340
57	560	750	12	260	340
56	550	740	11	260	330
55	540	730	10	250	320
54	540	720	9	240	310
53	530	710	8	230	310
52	520	700	7	230	300
51	520	690	6	220	290
50	510	680	5	210	280
49	500	670	4	210	280
48	500	660	3	200	270
47	490	650	2	200	260
46	490	640	1	200	250
45	480	640	0	200	250
44	470	630	– 1	200	240
43	470	620	– 2	200	230
42	460	610	– 3	200	230
41	450	600	– 4	200	220
			– 5	200	210
			– 6 or below	200	200